

WHAT A LANGUAGE MODEL DOES WITH A SOLVER CERTIFICATE: AN ANSWER-EVIDENCE DECOMPOSITION ACROSS FOURTEEN MODELS

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ABSTRACT

Solver-, tool- and retrieval-augmented systems report large accuracy gains, routinely read as evidence that the model used the supplied material. Two mechanisms give the same number: reading a supplied answer, and reasoning over supplied state. The standard control, a same-shape irrelevant record, separates neither. We build one that does, holding every surface property we could enumerate and audit fixed while destroying validity alone, across four sealed experiments and three robustness checks: 38,424 scored responses, re-derivable from released receipts. Asked what a model does with a certificate, the instrument returns a different answer for almost every checkpoint, arm and inference runtime. Breaking validity alone removes 98.3% of the state effect on one checkpoint and 40.4–51.4% on another; inference depth is one step on two checkpoints and not on a third; one model is a usable interface under one inference stack and below the viability floor under another, on identical weights. One result does not move. Given a certificate whose fabricated final rule establishes the query’s negation, all three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, and the strongest abstains on 0 of 192 broken certificates. Mechanism claims about solver assistance do not transfer between checkpoints. The failure to check the solver does.

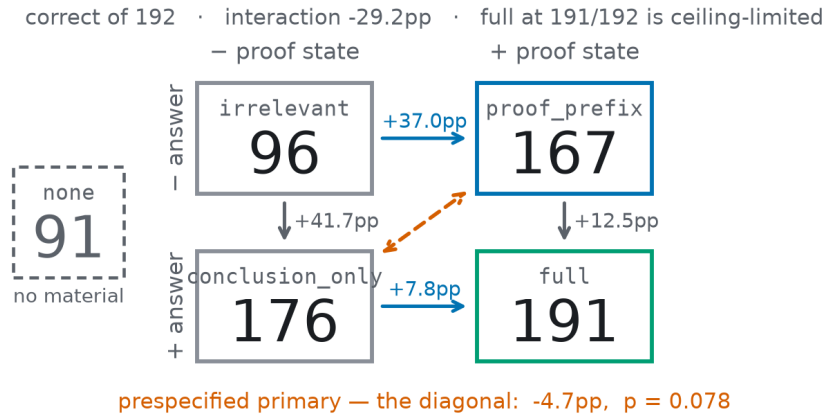


Figure 1: The instrument. A solver certificate carries an answer and the state it follows from, and accuracy adds the two together. Across by whether the record carries the answer, down by whether it carries the state: every arrow is then a one-factor edge that attributes. Cells give correct items of 192 for the reanalysed experiment. The dashed diagonal moves both factors at once, which is why its null attributed to neither.

1 INTRODUCTION

Pairing a language model with an external solver, tool or retrieval index reliably raises accuracy on tasks the model cannot do alone. Systems that translate a problem into a formal representation, dispatch it to a symbolic engine and return the verdict through the model report large end-to-end gains (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2023; Gao et al., 2022; Schick et al., 2023; Yang et al., 2023; Lewis et al., 2020). Those gains are routinely read as evidence that the model *used* the supplied material.

That reading is underdetermined. Under *answer reading* the supplied record contains the conclusion and the model copies it out; under *state use* the model works with the supplied premises and completes the inference itself. Both give the same accuracy. The standard control, a same-shape but irrelevant record, rules out proof length, formatting and generic validity, but not answer reading: a prefix stopping one step short still names the query’s subject and predicate in adjacent lines, so a model that only checks co-occurrence reproduces every number a reasoning model would.

The distinction is not academic. A model completing a supplied inference has some purchase on whether that inference is sound; a model matching surface co-occurrence has none, and anything built on it inherits the solver’s errors in silence. No accuracy number separates the two.

Separating them needs a control that holds surface form fixed and destroys validity alone. We build one and apply it across four sealed experiments and three robustness checks, on frozen checkpoints from 0.5B to 70.6B, scoring by direct next-token likelihood over verified single-token candidates so that no free text is parsed.

The instrument works, and almost nothing it returns transfers. Breaking validity alone removes 98.3% of the state effect on one checkpoint and 40.4–51.4% on another. Inference depth, order-sensitivity and whether a model can serve as the interface at all move the same way: they are properties of a checkpoint, an arm and an inference runtime, not of the architecture. We therefore offer the method and no general value, ours included. The boundary is strict, too: on the Qwen checkpoints every arm whose record never displays the query’s predicate returns $d' = 0$, so high accuracy with a full certificate describes a faithful transcriber rather than a reasoner.

One result does not move. Given a certificate whose fabricated final rule establishes the query’s negation, all three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, and the strongest abstains on 0 of 192 broken certificates. Mechanism claims about solver assistance do not transfer between checkpoints. The failure to check the solver does.

What is new is the instrument: a 2×2 separating solver-supplied answers from solver-supplied state (fig. 1), a surface-matched invalid certificate certified per item, and a closure certifier for queries nothing decides. We apply it to fourteen models, thirteen of which the analysis interprets (Qwen Team et al., 2024; Yang et al., 2025; Grattafiori et al., 2024; Gemma Team et al., 2025; Microsoft et al., 2025; OLMo Team et al., 2025; Allal et al., 2025).

2 RELATED WORK

Chain-of-thought made intermediate steps part of the output (Wei et al., 2022), and a literature then asked whether the answer rests on them: explanations omit the biasing features that drive the prediction (Turpin et al., 2023), injected mistakes are often ignored (Lanham et al., 2023; Arcuschin et al., 2025), answer-determining hints go unverbilised (Chen et al., 2025), and one response is to make the trace a program a solver executes (Lyu et al., 2023). Neurosymbolic work puts the formal step outside the model altogether (Pan et al., 2023; Olausson et al., 2023; Ye et al., 2023; Gao et al., 2022), and the pattern extends to tool use (Schick et al., 2023), premise selection (Yang et al., 2023) and retrieval (Lewis et al., 2020).

Both leave our question open. The first intervenes on *model-generated* traces, where ignoring a mistake is unfaithful; ours is externally supplied, formally verified state, where following a corrupted one is unsafe. The second reports end-to-end accuracy, which cannot separate the solver’s *answer* from its *state* and assumes the solver correct, though the model side is not neutral to context (Shi et al., 2023; Xie et al., 2023).

Table 1: What the model sees, from sealed item b15-4ef53c71. Every prompt is the same 18-line theory in invented vocabulary, then one solver record, then the query. Only the record changes between arms. Left: the FULL record, a complete four-step derivation. Right: what each other arm does to it. BROKEN_CHAIN is the word-matching test: line 6 becomes a different *real* rule of the theory, one that does not fire, so the chain no longer reaches the query while every audited surface cue of TRUNCATE_1 survives. MISLEADING instead invents a rule the theory does not contain.

Query: Bimol is vrinpek. (true)	
Arm	Record supplied
NONE	no record at all
IRRELEVANT	same shape, a different entity throughout
SAME_ENTITY_IRREL.	same entity, an unrelated predicate
TRUNCATE_1	lines 1–8: the answer line withheld
TRUNCATE_2	lines 1–6: two steps withheld
TRUNCATE_3	lines 1–4: three steps withheld
SHUFFLED	lines 1–8 in seeded random order
BROKEN_CHAIN	lines 1–8, but 6 becomes <i>All foskdem people are zuftbas.</i> and 7 becomes <i>Bimol is zuftbas.</i> Line 8 unchanged
MISLEADING	lines 1–9, but 8 becomes <i>All gasplaek people are not vrinpek.</i> and 9 becomes <i>Bimol is not vrinpek.</i>
FULL	lines 1–9

The task family descends from rule-based deductive datasets in natural language (Clark et al., 2020; Tafjord et al., 2020; Saparov & He, 2022; Han et al., 2022). Order effects there are large (Chen et al., 2024), the sets predate the models evaluated on them (Deng et al., 2024; Golchin & Surdeanu, 2023), and counterfactual variants show how much apparent competence rests on familiarity (Wu et al., 2023). None supplies the matched adversarial certificates an attribution question needs, so we generate items after model release with per-item invented vocabularies and build the certificates ourselves.

Prompted classification is biased rather than merely noisy: accuracy swings with format and example order (Zhao et al., 2021), option order alone moves it (Pezeshkpour & Hruschka, 2023; Zheng et al., 2023), likelihood scoring lets strings compete for probability mass independently of correctness (Holtzman et al., 2021), and abstention is its own capability (Kadavath et al., 2022; Yin et al., 2023). On a balanced panel a degenerate single-label responder is indistinguishable from a chance-level competent one, so we report d' and criterion c for every arm (Macmillan & Creelman, 1991; Stanislaw & Todorov, 1999) with the exact McNemar test (McNemar, 1947) and bootstrap intervals (Efron, 1987). Two cases here motivate that rather than illustrate it: an arm scoring 50.0% with $d' = 0.00$, and a prespecified criterion of our own passed by models answering one label almost everywhere.

3 METHOD

3.1 THE ANSWER-EVIDENCE DECOMPOSITION

A solver certificate carries an answer and the state it follows from, and fig. 1 separates them. IRRELEVANT supplies a valid derivation for a different entity, CONCLUSION_ONLY the derived terminal literal alone, PROOF_PREFIX the chain with that literal withheld, and FULL both, with a no-material baseline outside the square. The experiment we reanalyse chose the diagonal as its primary contrast.

Why the usual control is not enough. Take a theory in which *Kexil* is *dovrant*, every *dovrant* thing is *plesh*, and every *plesh* thing is *marn*, with the query *is Kexil marn?* The standard control gives a same-shape derivation about a different entity, fixing length, format and generic validity. But the arm it is compared against ends on *Kexil is plesh*, one line from a rule naming *marn*: the query’s subject and predicate sit in adjacent lines. A model that never takes the last step, and only registers that the two terms co-occur where a derivation is heading, reproduces every number a model completing it would. The square localises the gain to state; it cannot show the state was used as inference.

Table 2: The four experiments and three robustness checks. Sealed panels use per-item invented vocabularies, are mutually disjoint, and fix derivation depth at four. The public corpus validates the certifier only: no item from it is scored by any model here.

Panel	Items	Experiment	What it does
b14	192 binary	reanalysis	re-reads a five-arm factorial as the 2×2 (fig. 1)
b15	192 binary	validity control	ten matched arms (table 1) on three checkpoints
b16	192 three-class	model sweep	ten models $\times$ five arms, with an Unknown class
b17, b17b	b15 items	abstention rescoring	re-scores broken certificates, Unknown allowed
b18	b15 items	line-order check	eight permutations of each record
b16 on CUDA	as b16	runtime check	transformers , not mlx-lm ; two larger models
b15 at scale	b15 items	scale check	the same arms at 32.8B and 70.6B
ProofWriter OWA	23,240	certifier validation only (Tafjord et al., 2020)	

3.2 SURFACE-MATCHED INVALIDITY: THE WORD-MATCHING TEST

Table 1 shows one sealed item and every record we supply for it. Call the prefix that stops one step short `TRUNCATE_1`: the strongest arm that withholds the answer, and the one a surface-matching model can still exploit. `BROKEN_CHAIN` strips that exploit. We cut one intermediate link, so the displayed lines no longer reach the query, holding fixed against `TRUNCATE_1` the query-entity occurrence count (audited per item), the query predicate in the final rule, the line count and the fact/rule/literal sequence. Every rule displayed is real; exactly one derived literal is false; all 192 items are certified to leave the query undecided under the displayed lines. That gives

$$\begin{aligned} \text{total state} &= \text{TRUNCATE_1} - \text{IRRELEVANT}, & \text{surface} &= \text{BROKEN_CHAIN} - \text{IRRELEVANT}, \\ \text{validity} &= \text{TRUNCATE_1} - \text{BROKEN_CHAIN}. \end{aligned}$$

Five companion arms separate the rest, listed in table 1: `SAME_ENTITY_IRRELEVANT` isolates entity repetition, `SHUFFLED` isolates line order, and `TRUNCATE_2` and `TRUNCATE_3` give a depth ladder. `MIS-LEADING` is the corrupted-solver arm: one fabricated final rule, absent from the theory, establishes the query’s negation, so following the supplied record means answering incorrectly while the true answer is unchanged.

3.3 CERTIFYING THAT NOTHING DECIDES A QUERY

Showing a query is derivable is easy: exhibit the proof. Showing nothing decides it needs the theory’s forward closure saturated. Our certifier works under open-world semantics with explicit negation: a negative literal holds only if something derives it, never by failure to prove the positive. It agrees with a public corpus (Tafjord et al., 2020) on 23,240 of 23,240 questions, 10,440 of them undecided; passes nine adversarial theories probing cycles, round-cap overruns, derived negation and absent symbols; and agrees with an independently written reference on 192 of 192 items. No item enters a sealed panel uncertified.

4 EXPERIMENTAL SETUP

Every experiment asks a rule-chaining query over a small theory of facts and universally quantified rules; table 2 names the seven.

The abstention rescoring generates no panel. It rescores the validity control’s items with a third candidate available, changing only the instruction line and the candidate set on byte-identical inputs; its `NONE` arm fixes the abstention floor under that same instruction. The sweep’s 64 undecidable items have no terminal literal to supply, so those two cells instantiate the 2×2 less exactly than the rest. The three robustness checks reuse these panels: eight line permutations of each certificate, recording where the final rule and its premise land; the sweep rescored through **transformers** on CUDA, adding Llama-3.1-8B and Qwen2.5-14B; and the validity control’s own items on that same stack at Qwen3-32B (Yang et al., 2025) and Llama-3.3-70B, on rented H100s.

Responses are scored by direct next-token likelihood over verified single-token candidates: Yes/No on the binary panels, and Yes/No/Unknown wherever the panel carries an undecidable class — the

Table 3: Signal detection for the reanalysis; hits and false alarms out of 96 each. Apparent chance and zero sensitivity are not the same thing.

Arm	Hits	FA	d'	c
NONE	9	14	-0.25	+1.17
IRRELEVANT	9	9	0.00	+1.29
CONCLUSION_ONLY	80	0	+3.52	+0.81
PROOF_PREFIX	71	0	+3.20	+0.97
FULL	95	0	+4.72	+0.20

model sweep, the abstention rescoring and the runtime check. The model generates no free text, so no verdict is inferred from prose. Every prompt is instruction, theory, solver record and query, and the theory is visible in every arm. In the reanalysis the record’s header differs between arms as well as its content, a confound on that experiment’s primary contrast, removed everywhere after it.

Models. The reanalysis and validity control use one frozen 3B instruction-tuned checkpoint at 4-bit precision (Qwen Team et al., 2024), pinned to byte-identical weights. The sweep scores ten instruction-tuned models from 0.5B to 7.6B across six families (Qwen Team et al., 2024; Grattafiori et al., 2024; Gemma Team et al., 2025; Microsoft et al., 2025; OLMo Team et al., 2025; Allal et al., 2025) through `mlx-lm`, from pinned mirrors; the runner recorded no dtype, so we claim no uniform precision across the ten. No gate blocks that run: delivery is the capability under test, so a model that cannot emit a verdict reliably is a finding, not an exclusion.

Sensitivity, not accuracy. Accuracy on a balanced panel is uninterpretable when the responder is biased: one arm below scores 50.0% while discriminating nothing at all. We therefore report d' and criterion c for every arm (Macmillan & Creelman, 1991; Stanislaw & Todorov, 1999) using the loglinear correction, balanced accuracy for the three-class panel (chance 33.3%), the exact two-sided McNemar test (McNemar, 1947) as the decision criterion, and seeded paired bootstrap intervals (Efron, 1987) adjusted for multiple comparisons within declared families.

5 RESULTS

5.1 THE STATE EFFECT IS REAL, AND IT IS VALIDITY

The reanalysis begins from a null: its prespecified primary contrast, `PROOF_PREFIX` – `CONCLUSION_ONLY` = -4.7pp (exact McNemar $p = 0.078$), was reported as no effect. That contrast is the diagonal. The edges tell a different story: both channels carry real effect, and not additively. The state main effect is $+22.4\text{pp}$, the answer main effect $+27.1\text{pp}$, and the interaction -29.2pp , larger than either. All four edges of fig. 1 clear exact McNemar $p < 10^{-4}$; the original reported the two state edges as secondary and neither answer edge at all. This reading is post hoc, re-derived from the 960 raw receipts by a script that asserts it reproduces all five published per-arm counts before reporting. The design is ceiling-limited, `FULL` reaching 191/192, so both “other factor present” edges are compressed and the main effects deflated with them.

The baseline description was also wrong. `IRRELEVANT` returns an identical 9/96 “Yes” rate in both classes at $d' = 0.00$ (table 3), matching a trivial always-No strategy’s 96/192 exactly; `NONE` falls below it, at 91/192 and $d' = -0.25$.

Destroying validity alone removes almost all of the state effect. `TRUNCATE_1` – `BROKEN_CHAIN` = $+60.4\text{pp}$ (exact McNemar $p = 6.9 \times 10^{-18}$, BCa $[+49.0, +68.8]$, entailed subset $n = 96$ as declared before the run), with d' dropping 2.446. Neither arm produces a false alarm, so the whole collapse is in the hit rate, 59/96 to 1/96; a criterion shift would have moved both. The total state effect of $+61.5\text{pp}$ is therefore $+1.0\text{pp}$ of surface overlap ($p = 1$) and $+60.4\text{pp}$ of validity — a **validity share of 98.3%**, 95% percentile $[94.4, 100.0]$ (table 4).

Two supporting contrasts. Entity repetition explains nothing ($+0.0\text{pp}$, $p = 1$) despite one arm naming the query subject four times and the other not at all. And depth is a cliff, not a slope, on the Qwen checkpoints: withholding two steps scores identically to supplying no certificate, 0/96. Gemma-3-4B is not in that position, and there the direction is worse than null — 4/96 against 22/96 unaided.

Table 4: The validity control: ten matched arms scored on three checkpoints, entailed subset (correct of 96, with signed d'). Only the solver record differs between arms. The share rows give the validity component as a fraction of the total state effect, against the IRRELEVANT control and against the unaided NONE arm; the two coincide wherever unaided competence is zero.

Arm	Qwen2.5-3B 4-bit		Qwen2.5-3B bf16		Gemma-3-4B	
	/96	d'	/96	d'	/96	d'
NONE	0	0.00	0	0.00	22	1.83
IRRELEVANT	0	0.00	0	0.00	2	0.62
SAME_ENTITY_IRRELEVANT	0	0.00	0	0.00	3	0.77
TRUNCATE_3	0	0.00	0	0.00	4	0.88
TRUNCATE_2	0	0.00	0	0.00	4	0.88
TRUNCATE_1	59	2.85	72	3.23	96	5.13
BROKEN_CHAIN	1	0.41	1	0.41	58	2.83
SHUFFLED	14	1.53	12	1.43	86	3.80
MISLEADING	0	-4.36	0	-4.06	0	-5.13
FULL	96	5.13	96	5.13	96	5.13
Validity share vs IRRELEVANT	98.3%		98.6%		40.4%	
Validity share vs NONE	98.3%		98.6%		51.4%	

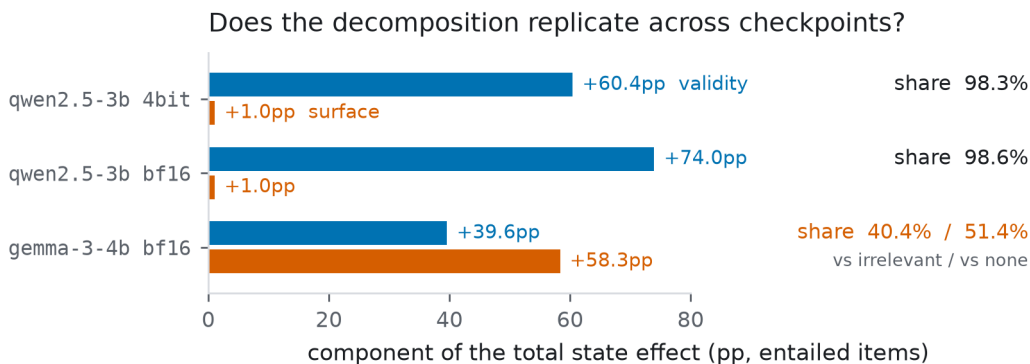


Figure 2: The surface and validity components on three checkpoints. Bars give the decomposition against the IRRELEVANT control; the annotation gives the share against that control and against what the model manages unaided. The two denominators coincide on the Qwen checkpoints, which score 0/96 with no certificate, and diverge only where a model has competence of its own.

5.2 ALMOST NOTHING THE INSTRUMENT RETURNS TRANSFERS

The instrument transfers; the number it returns does not. The decomposition was run on two further checkpoints: the same model at bfloat16, which isolates quantisation from model identity, and a different family. Figure 2 plots the split.

Quantisation is the smaller effect and leaves the share unchanged. Model identity is not like that. Gemma-3-4B answers BROKEN_CHAIN correctly on 58 of 96 items where the other two manage 1, so the control that isolates inference on one checkpoint exposes surface exploitation on another. Its share is 40.4% against the IRRELEVANT control and 51.4% against what it manages unaided; the divergence is the finding, not either number. The control is not inert on such a model either, costing Gemma most of its own competence, so any effect measured against it there is inflated. Nor does order-sensitivity replicate alongside the share (table 4).

The square itself is no steadier. Across ten models, post hoc from receipts already collected, the mean state main effect is +8.7pp against a mean answer main effect of +21.6pp: the answer channel dominates, and the reanalysis's +22.4pp is not typical. Excluding four responders degenerate in every arm gives +14.7pp against +28.5pp; the ordering survives either subset.

Table 5: All five arms, all twelve models. Each cell is balanced accuracy (% , chance = 33.3) / minimum per-class recall; bold clears the 0.50 recall floor, so VIABLE IN is derivable from the cells beside it. Shaded rows are the scale extension on a second runtime: the parameter ladder continues, the numbers are not comparable across the rule. Viability is reported per arm, not per model.

Model	B	NONE	IRREL	CONCL	PREFIX	FULL	VIABLE IN
Qwen2.5-0.5B	0.5	35.4/0.00	34.9/0.00	41.1/0.00	43.2/0.00	63.5/0.00	—
OLMo-2-1B	1.0	35.9/0.00	33.3/0.00	53.6/0.00	34.9/0.00	42.2/0.00	—
Llama-3.2-1B	1.2	33.9/0.00	33.9/0.00	57.8/0.00	36.5/0.00	37.0/0.00	—
Qwen2.5-1.5B	1.5	33.3/0.00	33.3/0.00	43.2/0.00	34.9/0.00	37.0/0.00	—
SmolLM2-1.7B	1.7	33.3/0.00	33.3/0.00	45.3/0.00	39.6/0.00	69.3/0.38	—
Qwen2.5-3B	3.0	41.1/0.00	39.1/0.00	96.4/0.91	74.0/0.56	90.6/0.72	concl, prefix, full
Llama-3.2-3B	3.2	45.8/0.00	43.2/0.02	67.2/0.02	55.2/0.02	80.7/0.42	—
Phi-4-mini	3.8	44.8/0.00	39.6/0.00	68.2/0.05	71.9/0.33	90.6/0.72	full
Gemma-3-4B	4.3	63.5/0.36	51.0/0.06	100.0/1.00	84.4/0.53	96.4/0.89	concl, prefix, full
Qwen2.5-7B	7.6	33.3/0.00	33.3/0.00	97.9/0.94	67.2/0.02	71.4/0.14	concl only
Llama-3.1-8B	8.0	38.5/0.00	34.4/0.00	65.6/0.00	64.6/0.00	66.7/0.00	—
Qwen2.5-14B	14.7	57.3/0.00	35.4/0.00	80.7/0.42	70.3/0.12	82.8/0.48	—

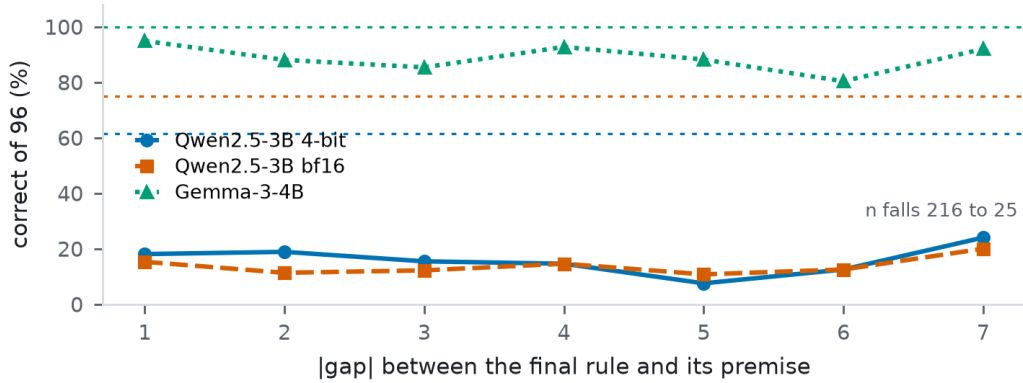


Figure 3: Accuracy against the gap between the final rule and the premise it fires on: all seven gaps, all three checkpoints, each checkpoint’s canonical-order score dashed. Distance is flat; direction is not, and the size of the direction effect is not shared.

Interface viability varies the same way. We had declared that a model relays non-determination at $\geq 80\%$ recall on undecidable items with full solver material. Eight of ten passed and the criterion was invalid: single-class recall is maximised by answering Unknown to everything, which four models approach. We replace it **post hoc** with a floor on every class, minimum per-class recall ≥ 0.50 , keeping both verdicts released. A second prespecified criterion, a paired FULL — NONE of 20pp on those items, is met by five of ten under Holm — two of them rejected by the replacement floor, the same gameability by another route (table 5).

Read by row: Qwen2.5-7B falls from second best in the sweep under CONCLUSION_ONLY to 0.141 minimum per-class recall under FULL: supplying the proof state is what breaks it. It lowers per-class recall on three of ten models and takes only this one below the floor. Nothing below 3B clears the floor in any arm, but that is a floor for *tolerating full certificates* rather than a size law: Llama-3.1-8B is degenerate in every arm, and Qwen2.5-14B never clears it.

The third factor is the inference stack. Re-scoring the same sealed panel, same weights, same prompt bytes through transformers on CUDA covers nine of the ten; eight agree on 95.5–99.7% of responses with no verdict moving. Phi-4-mini agrees on 87.9%, and its FULL arm is viable under one stack and not the other: minimum per-class recall 0.719 against 0.219 on identical weights, flips reaching 3.0 logits. Gemma-3-4B, which carries the 40.4–51.4% share, is the one model this check cannot cover, so that share has no runtime replication.

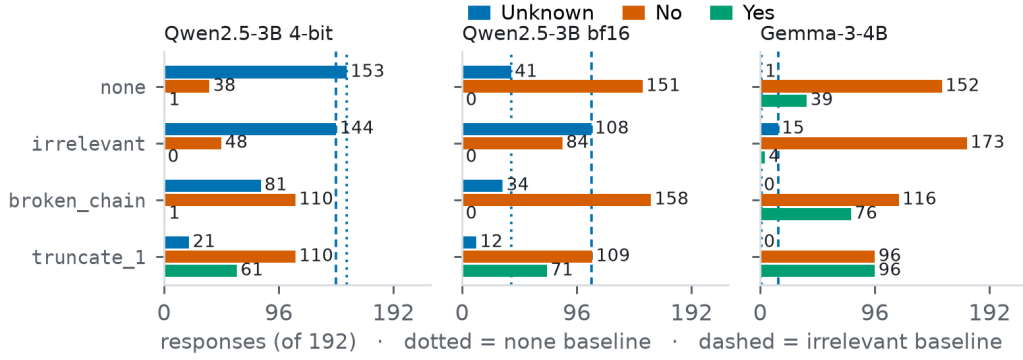


Figure 4: Three-candidate response distribution, three checkpoints, four arms. Dotted and dashed lines mark the NONE and IRRELEVANT abstention baselines, 4.7pp apart at 4-bit and 34.9pp at bfloat16. A broken chain lowers abstention against *both* references on every checkpoint, the opposite of the direction validity detection predicts.

5.3 WHAT DOES NOT VARY

Reversing the final rule and the premise it fires on costs accuracy on every checkpoint. Separating them does not. The split is post hoc: SHUFFLED moves three things at once, so we re-scored the same items under eight permutations each, recording where those two lines landed. An identity arm reproduces TRUNCATE_1’s bytes and returns its receipts 192/192 throughout. Distance is flat; direction is not (fig. 3). Rule *after* its premise gives 26.8, 22.4 and 94.9% across the three checkpoints against 5.8, 4.8 and 84.9% for rule *before*; paired within item on the 4-bit checkpoint, 45 items favour rule-after against 2 ($p = 1.6 \times 10^{-11}$). The direction holds everywhere, its size does not: a 4.6–4.7 $\times$ ratio on the Qwen checkpoints is a 10pp modulation on Gemma-3-4B.

Nothing defends against a wrong solver. All three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, at d' of -4.36 , -4.06 and -5.13 : the model exploiting surface overlap most is also the one most completely misled. The class split is sharper. The 4-bit checkpoint’s standing prior is No — under NONE it answers No to all 192 items, at $c = 2.565$ — and under MISLEADING it answers Yes on 93 of the 96 contradicted items. A fabricated rule inverts a near-total prior.

This does not attenuate with scale. Llama-3.3-70B, sixteen times the largest of the three above, scores 0 of 192 under MISLEADING at $d' = -5.13$, no item within 0.5 logits of the boundary. It is not struggling: 192/192 under FULL, 191/192 under BROKEN_CHAIN, and 145 of 192 with no certificate at all. A single fabricated rule inverts every one of them. Qwen3-32B is not interpretable here: it fails the per-class recall floor under FULL at 0.167 while clearing it on seven of the other nine arms, and the reason is measurement rather than the model — the pair we score disagrees with the token it would emit on 5 of 12 sampled items in that arm, against 12/12 where it is confident. We record it and draw nothing from it.

Whether any model notices, we cannot say. Rescoring BROKEN_CHAIN with an Unknown candidate available should raise abstention if the model tracks validity. It falls by 37.5 and 32.8 points against NONE and IRRELEVANT and *rises* by 31.2 against the surface-matched TRUNCATE_1 (fig. 4): the sign is a function of the reference, and neither reference is clean, so we draw no conclusion and withdraw the claim that it refutes detection. What needs no reference is that the most capable of the three abstains 0 times in 192 on a broken certificate and answers Yes on 76. No positive evidence, then, that any checkpoint verifies validity — and the corruption arm above is evidence that none does.

6 LIMITATIONS

Scope of the mechanism claim. The one-step depth reported above is a property of the Qwen checkpoints, as is the share: 40.4–98.6% against the IRRELEVANT control, 51.4–98.6% against unaided. The full theory is visible in every arm, so breaking a certificate is diagnostic only where the model scores 0/96 from the theory alone — true of both Qwen checkpoints, not of Gemma-3-4B, where

20 of its 22 unaided successes recur under `BROKEN_CHAIN`. A model using a broken certificate as a pointer back into the theory would be doing inference of a different kind; we record that as an interpretation caveat, not a controlled alternative. `TRUNCATE_2` and `TRUNCATE_3` also drop the query predicate, confounding depth with predicate presence, and an undecidable `PROOF_PREFIX` cannot contain it either.

Post hoc analyses and superseded criteria. Three analyses are post hoc and labelled where used, with the originals retained in the released results. The reanalysis carries five defects we report rather than repair: a prespecified 15pp target with 8.3pp of headroom above the 91.7% arm it applied to; no recorded proof depth, so its failures cannot be stratified; task identifiers encoding the semantic class, from which its answer key is reconstructed, there being no sealed panel; and a leak screen that was a malformed regular expression and never fired (the released certificates contain none, checked after the fact). Every later experiment fixes all five.

Candidate tokenisation. The prompt ends `Answer:`, inviting a space-prefixed continuation, while the scored candidates are the bare tokens. Only their relative order matters, and on the checkpoint carrying the scale claim the two pairs agree on all 36 sampled verdicts, diverging only near indifference — which is why Qwen3-32B is not interpreted. Two checkpoints are audited this way; the receipts bound the rest. On the validity control the median top-two margin is 6.9–11.3 logits with under 1% of responses within 0.5 of indifference; in the sweep it reaches 25.7–40.4% on three checkpoints — SmolLM2-1.7B, Llama-3.2-3B and Llama-3.1-8B — whose cells of table 5 are the least secure numbers here.

The scale check is one checkpoint. Llama-3.3-70B extends the corruption result past the small-model reading, but it is one model at that scale and its unaided competence leaves the validity share undefined: the denominator of section 3.2 collapses once a checkpoint can answer without the record. Its 191/192 under `BROKEN_CHAIN` is not evidence of detection either: a model with that much unaided competence may simply be ignoring a useless record. What the arm shows is narrower and harder to explain away: a fabricated rule overrides reasoning the model has.

Generalisation. One task family, small models plus one 70B scale check, synthetic panels. Invented vocabularies establish item novelty, not independence from every relevant pretraining pattern (Golchin & Surdeanu, 2023; Deng et al., 2024); model is not randomised, so cross-model comparisons are descriptive. The abstention rescoring changes the instruction and candidate set, and response distributions move substantially (Zhao et al., 2021; Zheng et al., 2023), so its contrasts hold and its rates do not. The reanalysis and validity control use a 4-bit checkpoint and the sweep bfloat16, different artefacts of one model; the sweep also chose the validity control’s checkpoints, leaving Phi-4-mini unrun on those arms. No deployed setup scores a forced choice among verified single-token candidates. At $n = 96$ the loglinear correction censors d' at ± 5.13 , where five cells of table 4 sit, so differences involving a saturated arm are bounds. The reanalysis’s per-arm counts and primary contrast appeared in an earlier unpublished report by the present authors, which concluded the 3B checkpoint failed an open-world unknown gate the sweep finds the family clears; differing panels, candidate sets and quantisation plausibly explain that, and we flag it rather than let a reader find both. Finally, every theory here arrives *already formalised*: nothing tests whether a model can turn a prose framework into solver input, which the corruption result makes consequential.

7 CONCLUSION

End-to-end accuracy cannot say why solver assistance helps. The decomposition can: arrange the arms so each contrast moves one factor, and break validity while holding surface form fixed. Across fourteen models, two inference stacks and 38,424 responses it returns a different answer almost every time it is asked: the validity share, inference depth, order-sensitivity and viability itself are properties of a checkpoint, an arm and a runtime. We offer the method and decline to generalise its value, ours included.

One result does not move. Given a certificate whose fabricated final rule establishes the negation, the three checkpoints answer incorrectly on 189, 186 and 192 of 192; nor is it a small-model artefact, since a 70.6B checkpoint that solves 145 of 192 unaided is wrong on all 192. Mechanism claims do not transfer between checkpoints; the failure to check the solver does. Run a corrupted arm, report per checkpoint, and report sensitivity beside accuracy.

REPRODUCIBILITY STATEMENT

Code and data. The supplementary material accompanying this submission is self-contained: sealed panels and answer authorities, all 38,424 per-response receipts, the generators, runners, audits and analysis scripts, and the forward-closure certifier with its validation against the public corpus. Every number in this paper can be re-derived from a fresh unpack with no network access, no model weights and no external packages; the audits use the Python standard library only, so verification does not depend on resolving an environment. Panels are hash-stamped, and the runners for the validity control and the model sweep refuse a panel whose hash does not match, while the others record the hash without enforcing it. The answer authority is opened after all receipts are written, by convention in the analysis order rather than by a gate in the code. Scoring takes the highest-likelihood candidate token; on an exact tie the first candidate in the declared order wins, which across the validity control’s 5,760 prospective responses occurs twice. Each receipt carries a SHA-256 of its own contents, and all 38,424 verified with zero failures. The count is not typed into the manuscript: it is produced by `scripts/census_receipts_v1.py` from the released receipts and substituted at compose time.

Checking one number takes a minute. `scripts/verify.py` recomputes each headline value and prints it beside the value this paper states. Nine of its eleven claims re-derive from the receipts and the sealed panels; the cross-model mean and the model table read released analysis files, which the listed scripts regenerate: `python3 scripts/verify.py validity-share` recomputes the 98.3% of section 3.2 and shows the four counts it came from. `--all` adds Gemma’s two baselines, the corruption counts at both scales, the detection direction on three checkpoints, the 2×2 state main effect, the cross-model mean, the response census, a recomputed digest for every one of the 38,424 receipts, all sixty cells of table 5, and the ten printed properties of the worked example in table 1. It uses the standard library only, so a reviewer needs no environment beyond Python. Model weights are not redistributed, but each run records the pinned revision and a hash of the weight files before those files are deleted. A permanent archive with a DOI will accompany the camera-ready version.

AI-USE STATEMENT

A large language model was used as a coding and drafting assistant throughout: generating experiment code, analysis scripts and figure code, and drafting this manuscript. All experimental design decisions, the interpretation of results, and the claim boundaries were made and verified by the authors. Every numeric value reported was produced by the released analysis scripts from sealed receipts, not written by a model; the released audits allow any reader to re-derive them independently.

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